Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building cloud-native applications, security-focused integrations, and developer tooling. Experienced in Java, Python, JavaScript, React, Node.js, REST APIs, Kubernetes, AWS, Azure, CI/CD, unit testing, DevSecOps, debugging, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python microservice APIs** and Azure SQL systems for 500+ users, optimizing indexed queries, schemas, and backend execution paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented 5,000+ monthly workflows through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving **monitoring**, recovery, maintainability, operational resilience, and systematic production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed **React interfaces and Flask APIs** handling 10,000+ daily requests, implementing secure ingestion, authentication, progress states, and operational controls while systematically debugging production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, **AWS RDS** persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed scalable **Python services and REST integrations** using object-oriented patterns, AWS Glue, S3, SQL, filtering, pagination, and caching, improving production response times 30% and processing efficiency 68%.
- Maintained 20+ automated unit and integration suites with **Jenkins**, PyTest, and Mockito, strengthening CI/CD validation, defect investigation, release verification, and reliable deployments across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular **React and Node.js** features across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, processing 2 million+ Azure SQL records through documented software interfaces.
- Automated Docker builds, functional tests, security checks, staged deployments, and production monitoring through **Jenkins and Azure**, reducing release time 50% while improving reliability, deployment consistency, and operational visibility.

Cyber Security and Data Mining **India**
Data Analyst – Research Fellowship *May 2023 – August 2023*
- Built a **malicious-URL detection pipeline** over 650,000+ URLs using 21 behavioral indicators and classical machine-learning models, identifying phishing and malware patterns while achieving 96%+ classification accuracy.
- Evaluated Random Forest, Gaussian Naive Bayes, RNN, LSTM, Bi-LSTM, GRU, and CNN approaches, documenting model behavior and validating detection quality across diverse production-like security-focused test datasets.

Skills

Languages & Full Stack: Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3, React, Node.js, Flask, FastAPI, object-oriented programming
APIs & Data: REST APIs, microservices, WebSockets, PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, AWS RDS, relational databases, schema design, query optimization, caching
Cloud & DevSecOps: Kubernetes, Docker, AWS, Azure, GCP, S3, Jenkins, CI/CD, Git, GitHub, pull requests, code reviews, authentication, role-based access, secure coding, security testing
Quality & Delivery: unit testing, integration testing, functional testing, PyTest, Mockito, debugging, monitoring, root-cause analysis, SDLC, cloud-native development, deployment automation, documentation, production support, Agile

Projects

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a scalable platform supporting 50,000+ users and one million transactions, integrating **secure APIs**, token-based access controls, real-time WebSockets, MongoDB persistence, Kubernetes deployment, and automated release workflows.

Big-Brain | Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*
- Built secure document upload, authentication, access controls, semantic search, and collaborative workflows using **LLaMA-2, PostgreSQL, and REST APIs**, reducing document-review time 60% across 50,000+ indexed records.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an AI-powered database workspace supporting 10+ systems with schema-aware query generation, structured validation, optimized execution, and **Docker-backed deployment** across modular Java and TypeScript services in enterprise-style environments.